A Multi-vision Contextual Attention Approach for Vietnamese Visual Question Answering

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Abstract. Visual Question Answering (VQA) is a representative task that lies at the intersection of Computer Vision and Natural Language Processing. This task not only has significant scientific value but also offers considerable practical applications. For instance, integrating VQA models into mobile devices can assist blind and visually impaired individuals in accessing and understanding visual content. A common approach in existing VQA methods is to extract features from individual regions of an image, enabling models to capture local contextual information. However, such approaches often overlook global context, which can limit the model's ability to effectively aggregate information and perform multimodal reasoning. Recent studies have employed Vision Transformers to extract both global and local visual features, leading to improved model performance. In addition, multimodal attention mechanisms have been widely adopted to optimize the fusion process between visual and textual information, enabling models to better understand contextual relationships and focus on relevant features. Currently, many VQA models have been optimized for English datasets, while several approaches tailored for Vietnamese, known as Vietnamese Visual Question Answering (ViVQA), have also been proposed. In this paper, we propose an improved model for the ViVQA task based on the Multi-vision Contextual Attention architecture. Specifically, we introduce the Unified Attention Encoder, a novel mechanism designed to enhance the integration and relational modeling between image features and question representations through multi-context attention. In addition, we propose Hybrid Sequential Fusion, a new feature fusion strategy that integrates attention mechanisms with an LSTM-based sequential model to effectively aggregate multimodal representations before answer prediction. Experimental results on the ViVQA dataset demonstrate that the proposed approach achieves an accuracy of 72.15%, significantly outperforming the baseline model, which achieves 60%, thereby validating the effectiveness of the proposed components in improving multimodal reasoning capability.

Keywords: multimodal · natural language · PhoBERT · ResNet · Vietnamese visual question answering · EfficientNet · BLIP-2.

1 Introduction

In recent years, two important fields of artificial intelligence, Natural Language Processing (NLP) and Computer Vision (CV), have achieved significant progress [1]. While NLP focuses on enabling computers to understand and process human language, CV aims to analyze and interpret information from images. However, many real-world tasks require AI systems to simultaneously understand both language and visual information. To address this need, multimodal learning [2] [3] [4] [5] has emerged as a research direction that focuses on developing models capable of receiving, processing, and integrating information from multiple data modalities, such as text, images, audio, or video. As a result, these models can perform complex tasks that require the combination of different types of data.

A considerable number of tasks and models have been studied in this direction. For example, Image Captioning [6] [7] focuses on automatically generating natural language descriptions for images, while Text-to-Image Generation [8] [9] aims to generate images based on textual descriptions. In addition to these tasks, many multimodal models have been developed to learn joint representations between images and language. Notable examples include CLIP [10], which learns the correlation between images and text through contrastive learning [11], and BLIP [12], a model designed to improve the understanding and generation of content across the two modalities. Besides the tasks mentioned above, there exists another challenging multimodal task known as Visual Question Answering (VQA) [13]. Besides the tasks mentioned above, Visual Question Answering (VQA) is another challenging multimodal task that requires a model to answer natural language questions based on a given image, demanding a joint understanding of both visual and textual modalities. A typical VQA framework consists of three main stages: (1) extracting feature representations from the image and the question, (2) learning cross-modal interactions through a fusion mechanism, and (3) predicting the final answer. An overview of this general pipeline is illustrated in Figure. 1.

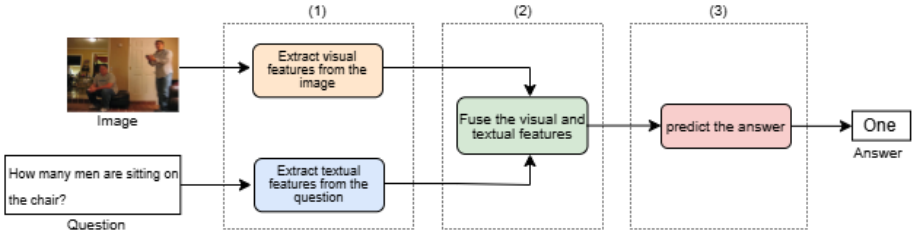


Figure 1: Overview of a typical Visual Question Answering (VQA) pipeline.

In the field of Visual Question Answering (VQA), English datasets have been extensively developed and play an important role in advancing research. Many large-scale datasets have been introduced, such as VQA v2.0 [14], which contains

hundreds of thousands of images and more than one million question–answer pairs, and GQA [15], which was designed to evaluate complex semantic reasoning based on visual information. These datasets are not only large in scale but also diverse in terms of question types, enabling models to learn various reasoning patterns. In contrast, VQA datasets for Vietnamese remain relatively limited. The number of publicly available datasets is still small, and their scale is significantly smaller compared to English datasets. This limitation poses challenges for training and evaluating VQA models in the Vietnamese language context. In this study, we use the ViVQA dataset, proposed by Tran et al [16], as the evaluation dataset for our model. This dataset is one of the representative benchmarks for Vietnamese Visual Question Answering and consists of image–question–answer pairs constructed based on images from the MS COCO dataset. Using the ViVQA dataset provides a standardized benchmark for training and evaluating our model in the Vietnamese VQA setting and allows comparison with previous studies that used the same dataset.

The scarcity of datasets not only makes it difficult to train large models but also limits the number of studies and models specifically developed for Vietnamese. Nevertheless, several VQA models for Vietnamese have been proposed in recent studies, such as the works presented in [17] [18] [19]. The main objective of our paper is to develop an effective VQA model for Vietnamese that can better exploit the relationship between visual information and natural language questions. Specifically, our model is developed based on the baseline model proposed by Duc Anh et al. [20], where we introduce several architectural improvements to enhance multimodal representation and feature integration.

In this study, we propose a VQA model for Vietnamese to improve the integration between image information and question representations based on the baseline model. Specifically, the baseline model utilizes the Guide Attention module as a mechanism for combining visual and textual features. In addition, we introduce a second fusion mechanism through the Unified Attention Encoder, which aims to better capture contextual relationships between the two modalities. The two multimodal feature streams are then integrated through the Hybrid Sequential Fusion module, which is designed based on the idea of combining the attention mechanism with an LSTM network to enhance the representation of joint features. This approach enables the model to effectively exploit multiple forms of interaction between images and questions in the Vietnamese VQA task.

The main contributions of this study are summarized as follows:

- We propose an improved model for Vietnamese Visual Question Answering (VQA), built upon the baseline model of Duc Anh et al., to enhance the interaction between visual and textual features.
- We introduce a dual multimodal fusion strategy, consisting of Guide Attention and Unified Attention Encoder, to effectively capture contextual relationships between visual features and question representations.
- We design a Hybrid Sequential Fusion module, inspired by the integration of attention mechanisms and LSTM networks, to combine multimodal feature streams and improve the representation of joint information.

- Experimental results on the ViVQA dataset demonstrate that the proposed method is effective and outperforms previous approaches.

2 Related works

2.1 Vietnamese Visual Question Answering datasets

In recent years, along with the rapid development of the Visual Question Answering (VQA) task, numerous datasets have been constructed to facilitate the training and evaluation of VQA models. However, the majority of these datasets have primarily been developed for English, while resources for Vietnamese remain relatively limited. One of the earliest efforts to build a Vietnamese VQA dataset is ViVQA [16], proposed by Tran et al. This dataset consists of 10,328 images and approximately 15,000 Vietnamese question–answer pairs, constructed based on images from the MS COCO [21] dataset. The questions in the dataset are designed to cover various aspects of images, such as objects, quantities, colors, and actions, in order to evaluate the model's capability in visual understanding and reasoning. ViVQA is currently considered one of the foundational datasets for Vietnamese VQA research and provides a benchmark for evaluating methods proposed in this field. To provide a clearer intuition of the dataset, several representative examples from ViVQA are illustrated in Figure. 2.

The OpenViVQA [22] dataset was proposed to provide a larger-scale VQA resource for Vietnamese and to address several limitations of previous datasets. This dataset includes 11,199 images and approximately 37,914 question–answer pairs, collected and annotated through a manual labeling process. The questions in OpenViVQA follow an open-ended format, allowing answers to take the form of a word, phrase, or complete sentence rather than being restricted to a predefined set of candidate answers. In addition, the dataset is divided into two main categories: Text QA and Non-text QA, where some questions require the model to extract textual information appearing within the images. The images in OpenViVQA are primarily collected from real-life contexts in Vietnam, reflecting local cultural characteristics and environmental settings, thereby creating question–answer scenarios that are close to real-world contexts. Thanks to this characteristic, the dataset not only provides a new benchmark for Vietnamese VQA models but also introduces additional challenges for existing approaches, particularly in tasks that require reasoning ability and natural answer generation.

The ViCLEVR [23] dataset was proposed to support research on visual reasoning in the context of Vietnamese VQA. This dataset contains more than 26,000 images and approximately 30,000 question–answer pairs, where each question is annotated with a reasoning type to evaluate the model's capability in understanding and logical reasoning. The images in ViCLEVR are generated as synthetic images with multiple objects, colors, and spatial positions, enabling the creation of questions that require reasoning about spatial relationships, quantities, or object attributes. Through this controlled design, ViCLEVR provides a



Question: Những gì đang đậu trên cành của cây? (What is perched on the branches of the tree?)
Answer: Con chim (Bird)



Question: có bao nhiêu người đàn ông đang ngồi trên ghế? (How many men are sitting in the chairs?)
Answer: một (One)



Question: màu của máy tính xách tay là gì? (What color is the laptop?)
Answer: màu cam (orange)



Question: đoàn tàu dừng ở đâu? (Where does the train stop?)
Answer: trạm (station)

Figure 2: Representative examples from the ViVQA dataset.

clear evaluation environment for Vietnamese VQA models, particularly for tasks requiring complex reasoning between visual information and questions.

The EVJVQA [24] dataset was proposed to expand the available resources for the Visual Question Answering task in a multilingual setting, particularly between English and Vietnamese. This dataset includes bilingual image–question–answer pairs, allowing models to learn and evaluate their ability to understand visual information combined with language in both languages. The images in EVJVQA are mainly collected from widely used datasets such as MS COCO, after which corresponding Vietnamese questions and answers are constructed alongside the English versions. Due to its bilingual nature, EVJVQA not only supports research in Vietnamese VQA but also facilitates studies on cross-lingual knowledge transfer [25] in VQA models.

The ViOCRVQA [26] dataset was developed to support research on Visual Question Answering combined with optical character recognition (OCR-VQA) [27] for Vietnamese. This dataset consists of images containing textual information that appears directly within the images, along with Vietnamese question–answer pairs that require models to extract and utilize such textual content to produce correct answers. The questions in this dataset often involve reading, understanding, and reasoning over textual information in the images, such as information on traffic signs, product labels, or public notices. Consequently, ViOCRVQA introduces a new challenge for Vietnamese VQA models,

as the models must not only understand visual content but also integrate text recognition and processing capabilities.

The ViTextVQA [28] dataset was constructed to support research on Visual Question Answering tasks involving textual content in images for Vietnamese. This dataset contains images with abundant textual information appearing in real-world contexts, together with Vietnamese question–answer pairs that require models to utilize the textual content to generate accurate answers. The questions often involve reading and understanding text in images, such as information on signboards, advertisements, or product labels. Due to this characteristic, ViTextVQA provides a challenging benchmark for Vietnamese VQA models, as the models must integrate both visual understanding and optical character recognition (OCR) capabilities to effectively address the task.

2.2 Representative Visual Question Answering Methods

In recent years, the rapid development of Large Language Models [29] (LLMs) and Vision–Language Models [30] (VLMs) has opened new directions for the Visual Question Answering (VQA) task. Owing to their strong natural language understanding capabilities and their ability to perform reasoning based on large-scale knowledge, these models enable image-related questions to be addressed in a more flexible and generalizable manner compared to traditional VQA architectures. Several recent studies have leveraged LLMs to enhance reasoning and explainability in VQA systems. For instance, Zhou et al [31]. proposed Code-Driven LLM Agent, in which an LLM is utilized as an agent capable of generating and executing code to support the reasoning process and produce explanations for answers in VQA tasks. In addition, Guo et al [32]. introduced RegionGPT, a vision–language model that enables LLMs to understand and process region-level information within images, thereby improving the ability to describe, reason about, and answer questions related to specific objects. Furthermore, Peng et al [33]. proposed KOSMOS-2, a large multimodal model capable of grounding language expressions to corresponding regions in images, enabling a direct connection between linguistic information and the visual world. Overall, these studies demonstrate that integrating LLMs into VQA systems is becoming an important trend, helping to improve reasoning capability, contextual understanding, and multimodal interaction. However, the application of such large-scale models still faces several challenges, including high computational costs, the requirement for large-scale training data, and limited adaptability to low-resource languages such as Vietnamese.

In recent years, in addition to leveraging large language models, another approach that has attracted considerable attention in the Visual Question Answering (VQA) task is the use of agent-based models or multi-agent systems [34]. These approaches treat the question-answering process as a sequence of reasoning steps carried out by multiple agents, where each agent is responsible for a specific role in analyzing the image, understanding the question, and producing a more accurate answer. Wang et al [35]. proposed a multi-agent VQA model capable of top-down reasoning, in which agents are designed to perform different

reasoning steps and collaborate with each other to generate clear explanations for the final answers. This approach improves the transparency and interpretability of the model during the process of answering questions based on images. In addition, Yu et al [36]. introduced the Dual Latent Memory model for Visual Multi-Agent systems, where agents share and update information through two latent memory modules to enhance collaboration and information exchange during the reasoning process. This mechanism enables the system to better exploit visual information and question context when solving VQA tasks. Overall, recent studies indicate that the application of agent-based models and multi-agent systems has the potential to improve the reasoning capability and flexibility of VQA models. Nevertheless, this approach still faces several challenges, such as the complexity of system design, high computational costs, and the difficulty of optimizing coordination among agents.

In the paper "Advancing Vietnamese Visual Question Answering with Transformer and Convolutional Integration" [37], the authors focus on improving the performance of Vietnamese VQA models by enhancing the feature fusion mechanism between images and text. Specifically, the model employs a multimodal architecture that enables deep interaction between image features and question representations, in which attention mechanisms play a crucial role in learning semantic relationships between the two modalities. In addition, the integration of convolutional features provides detailed visual information, supporting the model in scenarios that require recognizing small but important visual characteristics. This approach improves multimodal representation capability and enhances the model's reasoning performance. Experimental results on the ViVQA dataset show that the proposed method achieves an accuracy of 71.04% on the test set, outperforming previous baseline models.

In the paper "Combining Multi-vision Embedding in Contextual Attention for Vietnamese Visual Question Answering" [1], the authors propose a method for Vietnamese Visual Question Answering (VQA) based on the Guided Attention mechanism to enhance the interaction between image features and questions. Specifically, the question representation is used as a guiding signal to adjust the process of feature extraction and the distribution of attention weights over image regions, enabling the model to focus on areas that are directly relevant to the question content. This mechanism allows the model to learn multimodal relationships more effectively compared to traditional fusion methods. The model is evaluated on the ViVQA dataset and achieves an accuracy of 60.76%, showing a significant improvement over previous approaches and demonstrating the effectiveness of guided attention in Vietnamese VQA tasks.

3 Methodology

In this work, we formulate the Vietnamese Visual Question Answering (ViVQA) task as a classification problem. Given an input image I and a corresponding question T, the objective is to select the most appropriate answer from a predefined answer set A. Formally, the predicted answer is determined by maximizing

the conditional probability over the candidate answers.

$$\hat{a} = \arg\max_{a \in A} P(a \mid I, T) \tag{1}$$

This section presents the architecture of the proposed ViVQA model and its key components for multimodal reasoning. The model adopts the Guided Attention module from the baseline to facilitate interactions between visual and textual features. In addition, two modules, namely the Unified Attention Encoder and the Hybrid Sequential Fusion, are introduced to enhance contextual representation and multimodal fusion. The fused representation is then fed into a classifier to generate the final answer. The overall architecture is illustrated in Figure. 3.

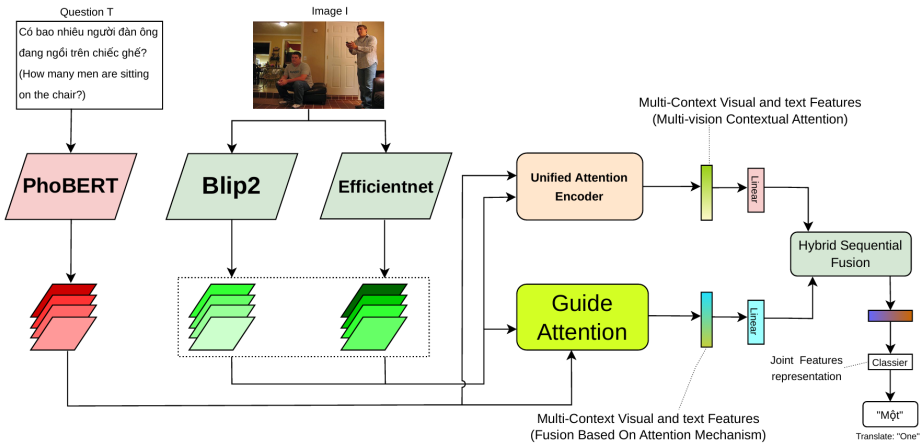


Figure 3: Overview of our proposed Multi-vision Multi-level Fusion model.

In the proposed model, we employ pretrained backbones to effectively extract multimodal features. Specifically, for the visual modality, we combine both local and global representations to leverage their complementary strengths: local features capture fine-grained details (e.g., objects and regions of interest), while global features provide the overall contextual information of the image. Accordingly, local features are extracted using EfficientNet-B7 [38], a variant of the EfficientNet family known for its strong and efficient visual representation learning, whereas global features are obtained from BLIP-2 [39], which enables high-level semantic understanding from multimodal data.For the textual modality, the question is encoded using PhoBERT [40], a pretrained language model for Vietnamese built upon the BERT architecture. As these backbones have been extensively studied in prior work, we do not elaborate on their architectures, but instead focus on their integration within the proposed framework. The corresponding multimodal representations are obtained from the pretrained backbones as follows:

$$V_{local} = \text{EfficientNet}-\text{B7}(I) \tag{2}$$

$$V_{global} = \text{BLIP}-2(I) \tag{3}$$

$$V_{text} = \text{PhoBERT}(T) \tag{4}$$

where V_{local}, V_{global}, and V_{text} denote the local visual features, global visual features, and textual features, respectively, which are used as inputs for subsequent multimodal fusion and reasoning modules.

3.1 Unified Attention Encoder

The Unified Attention Encoder branch is designed to learn multimodal representations at the global semantic level, complementing the Guided Attention baseline branch. The processing pipeline consists of three main stages: feature projection, multi-head attention encoding, and refinement via stacked self-attention.

3.1.1 Feature Projection into a Common Space

Visual features are extracted from two sources: global features from BLIP-2 ($V_{global} \in R^{768}$) and local features from EfficientNet ($V_{local} \in R^{768}$). These two vectors are concatenated to form a joint visual representation, which is then linearly projected into a $d = 1024$-dimensional space:

$$V_{CC_visual} = W_v[V_{global} \| V_{local}] + b_v, \quad V_{CC_visual} \in R^{1024} \tag{5}$$

In parallel, the question feature $V_{local} \in R^{768}$ obtained from PhoBERT is linearly projected into the same space:

$$V_{text} = W_q V_{text} + b_q, \quad V_{text} \in R^{1024} \tag{6}$$

where $W_v \in R^{1536\times1024}$ and $W_q \in R^{768\times1024}$ are learnable weight matrices.

3.1.2 Multi-Head Self-Attention

The two vectors V_{CC_visual} and V_{text} are concatenated to form a unified multimodal representation vector:

$$\mathbf{z} = [V_{CC_visual} \| V_{text}] \in R^{2048} \tag{7}$$

The vector $\mathbf{z}$ is then fed into a Multi-Head Self-Attention block with $H = 8$ attention heads, key dimensionality $d_k = 256$, and value dimensionality $d_v = 256$:

$$head_h = Attention\left(\mathbf{z}\mathbf{W}_h^Q,\ \mathbf{z}\mathbf{W}_h^K,\ \mathbf{z}\mathbf{W}_h^V\right) \tag{8}$$

$$Attention(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = softmax\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{9}$$

$$\mathbf{z}' = Concat(head_1, \ldots, head_H)\,\mathbf{W}^O \in R^{2048} \tag{10}$$

This mechanism enables the model to simultaneously capture cross-modal interactions across multiple representation subspaces, where each attention head focuses on a distinct semantic aspect of the relationship between the visual and textual modalities.

3.1.3 Stacked Self-Attention

The output $\mathbf{z}' \in R^{2048}$ is subsequently passed through $N = 2$ stacked Self-Attention layers to model higher-order semantic dependencies:

$$\mathbf{z}^{(0)} = \mathbf{z}' \tag{11}$$

$$\mathbf{z}^{(l)} = SelfAttention\left(\mathbf{z}^{(l-1)}\right), \quad l = 1, 2 \tag{12}$$

The final output $\mathbf{z}^{(2)} \in R^{2048}$ is passed through a linear projection layer to obtain the representation $\mathbf{V}_{SSA} \in R^{1024}$:

$$\mathbf{V}_{SSA} = \mathbf{W}_r\,\mathbf{z}^{(2)} + \mathbf{b}_r \tag{13}$$

The vector $\mathbf{V}_{SSA}$ is subsequently combined with the output $\mathbf{V}_{GA}$ from the Guided Attention branch within the Hybrid Sequential Fusion module for answer classification.

3.2 Hybrid Sequential Fusion

To simultaneously leverage the complementary strengths of the two feature extraction branches, namely the *Baseline Guide Attention* and the proposed *Unified Attention Encoder*, we introduce a **Hybrid Sequential Fusion** module. This module is designed to flexibly and adaptively integrate multimodal representations. As illustrated in Fig. 4, we present the overall architecture of the proposed Hybrid Sequential Fusion module.

Input Representations: Specifically, given two parallel pipelines, we obtain:

- V_{SSA}: features from the proposed Unified Attention Encoder
- V_{GA}: features from the baseline Guide Attention branch

To ensure consistency in the representation space, both feature vectors are projected into a shared embedding space of dimension 1024 via linear transformations:

$$J_{\mathrm{UAE}} = \mathrm{Linear}(V_{\mathrm{SSA}}) \tag{14}$$

$$J_{\mathrm{GA}} = \mathrm{Linear}(V_{\mathrm{GA}}) \tag{15}$$

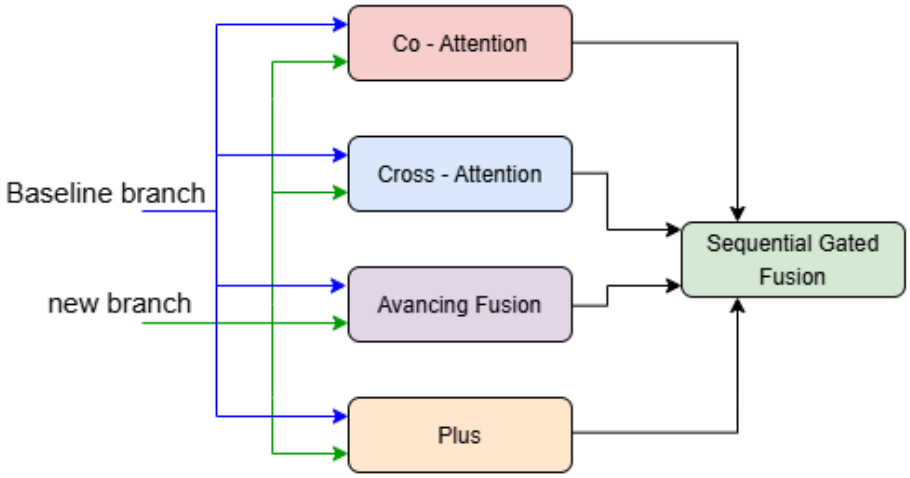


Figure 4: Overview of our proposed Hybrid Sequential Fusion module.

Although both representations encode joint information from the image and the question, each branch captures different interaction patterns. Therefore, effectively combining these representations is crucial for enhancing reasoning capability.

3.2.1 Multi-strategy Multimodal Fusion

The Hybrid Sequential Fusion module employs four complementary fusion strategies to exploit the interactions between J_{UAE} and J_{GA}:

In this context, B denotes the batch size, d_{UAE} and d_{GA} represent the number of tokens from the Guide Attention (baseline) branch and the Unified Attention Encoder (our proposed) branch, respectively, and d is the embedding dimension.

Additive Fusion. To combine information from the baseline branch (J_{GA}) and the proposed branch (J_{UAE}) in a simple yet effective manner, we employ element-wise addition between the two representations. Prior to this operation, both branches are projected into the same feature space and share the same dimensionality.

$$J_1 = J_{GA} + J_{\mathrm{UAE}} \tag{16}$$

Here, the addition is performed element-wise. This approach directly aggregates information from both sources while preserving the original representation structure without increasing computational complexity.

The final representation J_1 is then used for answer prediction.

Co-Attention Fusion. To exploit bidirectional interactions between the representations of the two branches, we employ a co-attention mechanism to learn a joint representation across modalities:

$$(J'_{GA}, J'_{\mathrm{UAE}}) = CoAtt(J_{GA}, J_{\mathrm{UAE}}) \tag{17}$$

Here, $J_{\text{UAE}} \in R^{B \times d_{\text{UAE}} \times d}$ and $J_{GA} \in R^{B \times d_{GA} \times d}$, where each CoAtt block performs mutual updates between the two modalities via a bidirectional attention mechanism. After multiple stacked layers, the model obtains semantically aligned representations between the two branches.

Next, the two features are fused through a fusion function. Specifically, we apply element-wise multiplication after aligning the two representations to the same dimensionality:

$$H = J'_{\text{UAE}} \odot Expand(Mean(J'_{GA})) \tag{18}$$

Finally, we apply an attention pooling mechanism over the feature sequence to obtain a global representation:

$$\alpha = softmax(\mathbf{W}H), \quad J_2 = \sum_i \alpha_i H_i \tag{19}$$

where $\mathbf{W}$ is a learnable parameter. The final representation J_2 is then used for answer prediction.

Cross-Attention Fusion. To model conditional dependencies between the two branches, we employ a cross-attention mechanism, where the representation of the first branch (baseline) is updated based on the representation of the second branch (our proposed). Unlike co-attention, which performs bidirectional updates, cross-attention allows one modality to act as the query while the other serves as contextual guidance.

$$J'_{GA} = CrossAtt(J_{GA}, J_{\text{UAE}}) \tag{20}$$

Here, $J_{UAE} \in R^{B \times d_{UAE} \times d}$ and $J_{GA} \in R^{B \times d_{GA} \times d}$. Each CrossAtt block performs attention with queries from the first branch and keys/values from the second branch, thereby enriching the representation of the first branch with semantic information from the second. This process is repeated across multiple layers to enhance alignment between the two branches.

After obtaining the conditionally updated representation J'_{GA}, we apply an attention pooling mechanism to extract a global representation:

$$\alpha = softmax(\mathbf{W}J'_{GA}), \quad J_3 = \sum_i \alpha_i J'_{GA,i} \tag{21}$$

where $\mathbf{W}$ is a learnable parameter. The final representation J_3 is then used for answer prediction.

Multimodal Fusion Module (MFM). Inspired by the multimodal fusion mechanism proposed in *Advancing Vietnamese Visual Question Answering* [37], we design a module to effectively integrate complementary representations from two distinct branches.

$$H = Concat(J_{GA}, J_{\text{UAE}}) \in R^{B \times (d_{GA} + d_{s\text{UAE}}) \times d} \tag{22}$$

To preserve modality-specific positional information, we employ a multiway positional embedding mechanism, which assigns different positional embeddings

to tokens from the two branches. The concatenated representation is then fed into a shared Transformer encoder:

$$Z = Encoder(H) \tag{23}$$

The encoder is implemented as a multiway Transformer, where the self-attention mechanism is shared across modalities, while modality-specific feed-forward layers are maintained separately.

$$J_4 = Z_{cls} = \tanh(\mathbf{W} \cdot Norm(Z_0)) \tag{24}$$

Here, Z_0 denotes the first token of the encoder output, $Norm(\cdot)$ is the layer normalization operation, and $\mathbf{W}$ is a learnable projection matrix. The final representation J_4 is then used for answer prediction.

These four strategies allow the model to capture diverse interaction patterns, ranging from simple linear combinations to complex contextual dependencies.

3.2.2 Sequential Gated Fusion

To leverage the complementary strengths of four different fusion strategies, we propose a hierarchical fusion mechanism based on gating to integrate the representations J_1, J_2, J_3, and J_4. This approach enables the model to adaptively balance the contributions of each vector. Our design is inspired by gating mechanisms in recurrent neural networks, which effectively control information flow.

Specifically, given two input vectors $J_a, J_b \in R^{B\times d}$, we define a binary fusion function as follows:

$$f = \sigma(\mathbf{W}_f[J_a; J_b]) \tag{25}$$

$$fuse(J_a, J_b) = f \odot J_a + (1 - f) \odot J_b \tag{26}$$

where $[\cdot;\cdot]$ denotes vector concatenation, σ is the sigmoid function, and $\odot$ represents element-wise multiplication.

Based on this formulation, the overall fusion process is performed in a two-stage hierarchical manner:

- **Stage 1:** Pairwise fusion

$$J_{12} = fuse(J_1, J_2), \quad J_{34} = fuse(J_3, J_4) \tag{27}$$

- **Stage 2:** Fusion of intermediate representations

$$J_{final} = fuse(J_{12}, J_{34}) \tag{28}$$

This mechanism allows the model to automatically learn the contribution weights of each fusion strategy through gating, resulting in a final representation J_{final} that is both informative and adaptive to individual samples. The representation J_{final} is subsequently used for answer prediction.

3.3 Training strategy

3.3.1 Answer Prediction

Upon obtaining the final multimodal representation from the Hybrid Sequential Fusion module, the unified feature vector is passed through a classifier to predict the answer.

The classifier is constructed as a two-layer fully connected network with weight normalization, ReLU activation, and Dropout regularization to mitigate overfitting. Specifically, given an input vector $J_{\text{final}} \in R^{1024}$, the prediction process is formulated as follows:

$$\mathbf{h} = \mathit{Dropout}(\mathit{ReLU}(\mathbf{W}_1^{norm}\,\mathbf{f} + \mathbf{b}_1)), \quad \mathbf{h} \in R^{2048} \tag{29}$$

$$\hat{\mathbf{y}} = \mathbf{W}_2^{norm}\,\mathbf{h} + \mathbf{b}_2, \quad \hat{\mathbf{y}} \in R^{C} \tag{30}$$

where $\mathbf{W}_1^{norm} \in R^{1024\times 2048}$ and $\mathbf{W}_2^{norm} \in R^{2048\times C}$ are learnable weight matrices normalized via weight normalization, C denotes the number of answer classes with a value of 353, and $\hat{\mathbf{y}}$ is the output logit vector.

The predicted answer label is obtained by selecting the index corresponding to the maximum value in the logit vector:

$$\hat{c} = \arg \max_{c\in\{1,\ldots,C\}} \hat{y}_c \tag{31}$$

3.3.2 Loss Function

To train the model, the Cross-Entropy loss combined with label smoothing is employed. Rather than using hard one-hot labels, label smoothing redistributes a portion of the probability mass to the remaining classes according to the following formulation:

$$\tilde{y}_c = (1-\epsilon)\, y_c + \frac{\epsilon}{C} \tag{32}$$

where ϵ is the smoothing factor and C is the total number of answer classes. The loss function is defined as:

$$\mathcal{L} = -\sum_{c=1}^{C} \tilde{y}_c \log p_c, \quad p_c = \frac{\exp(\hat{y}_c)}{\sum_{j=1}^{C}\exp(\hat{y}_j)} \tag{33}$$

The loss is aggregated over all samples in the batch (`reduction='sum'`). The incorporation of label smoothing discourages the model from becoming overconfident in its predictions, thereby improving generalization performance on the test set.

4 Experiment

The optimization process was carried out using the AdamW algorithm with hyperparameters $\beta = (0.9, 0.999)$. The model was trained for 25 epochs with a batch size of 32. The initial learning rate was set to 1×10^{-4}, combined with a weight decay of 1×10^{-5} to mitigate overfitting. Furthermore, a dropout rate of 0.3 was applied throughout the model layers to enhance generalization capability. Regarding the input data, images were resized to 224×224 pixels, while the maximum question length was limited to 20 tokens. All training procedures were conducted on a local server equipped with an NVIDIA GeForce RTX 5090 GPU to ensure computational efficiency.

4.1 Dataset and Evaluation Criteria

Dataset: In this study, we reuse the ViVQA dataset [16] to evaluate the proposed model. The dataset consists of 10,328 images and 15,000 Vietnamese question–answer pairs, constructed based on images from the MS COCO dataset. It is split into training and test sets with a ratio of 8:2, ensuring diversity in both visual contexts and question types for the VQA task.

Evaluation Criteria: Model performance is evaluated using Accuracy, defined as follows:

$$Accuracy = \frac{Numberofcorrectpredictions}{Totalnumberofsamples} \tag{34}$$

This metric measures the proportion of correctly predicted answers compared to ground-truth labels.

4.2 Results

4.2.1 Visualization of Training Process

Figure. 5 illustrates the training and test accuracy curves over epochs. As observed, the training accuracy increases rapidly during the initial epochs and approaches near-perfect performance after approximately 10 epochs, indicating that the model is able to effectively fit the training data. In contrast, the test accuracy improves steadily in the early stage but gradually stabilizes around 0.61–0.62, suggesting a performance plateau on unseen data.

The gap between training and test accuracy becomes more pronounced as training progresses, which indicates potential overfitting. Despite this, the test performance remains relatively stable after convergence, demonstrating that the model maintains consistent generalization ability. These results highlight the importance of incorporating regularization strategies or more effective fusion mechanisms to further improve generalization performance in VQA tasks.

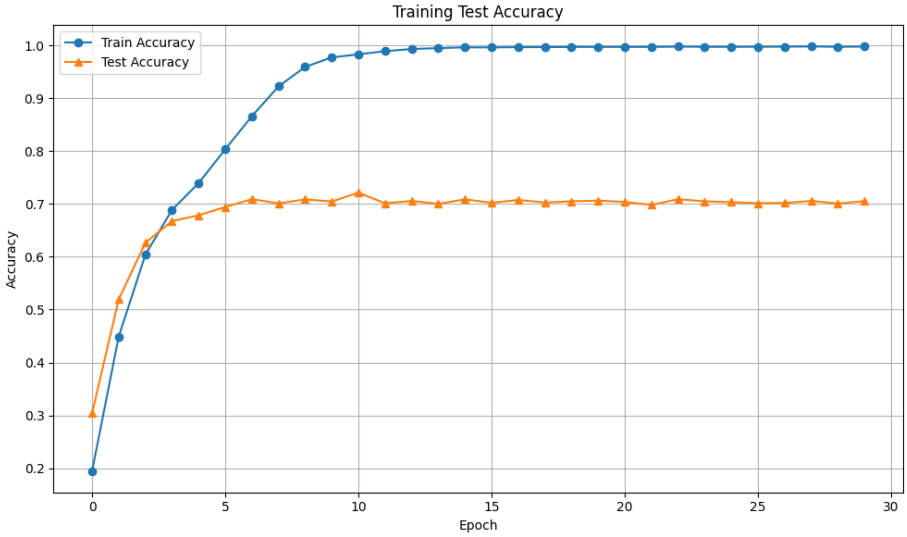


Figure 5: Training and test performance over epochs.

4.2.2 Comparison with Previous Studies

To evaluate the effectiveness of the proposed model, we compare it with several representative approaches in the VQA task. As shown in Table 1, traditional methods such as LTSM + PhoW2Vec, Bi-LTSM + PhoW2Vec, and HieCoAtt + PhoW2Vec achieve accuracies below 35%, indicating limitations in capturing multimodal relationships. More advanced approaches, including PAT, MCA, and BARTPhoBiT, significantly improve performance, achieving results from around 60% to nearly 69%.

Notably, the current model attains an accuracy of 71.04%, demonstrating substantial progress over prior methods. However, our proposed model further improves this result, achieving 72.15%, thereby surpassing the baseline and establishing the best performance among the compared approaches. This demonstrates the effectiveness of the proposed architecture in jointly exploiting visual and textual information more efficiently.

4.2.3 Comparison of Question Representation Backbones

To evaluate the impact of different language backbones for question representation, we conduct experiments using several widely adopted models, including BARTPho, ViT5, and PhoBERT. The results are summarized in Table 2.

As observed, both BARTPho and ViT5 achieve competitive performance, with accuracies of 68.59% and 69.24%, respectively. However, PhoBERT significantly outperforms the other backbones, achieving the highest accuracy of 72.15%. This result indicates that PhoBERT is more effective in capturing the semantic representation of questions, thereby substantially improving the overall performance of the VQA model.

Table 1: Comparison with previous Vietnamese VQA methods

Year	Question Backbone	Visual Backbone	Model	Accuracy (%)
2021	PhoW2Vec + LSTM	Linear transformation	LSTM + PhoW2Vec [16]	33.85
2021	PhoW2Vec + Bi-LSTM	Size normalization	Bi-LSTM + PhoW2Vec [16]	33.97
2021	PhoW2Vec + CNN + LSTM	4 Conv2D layers	HieCoAtt + PhoW2Vec [16]	34.96
2023	FastText	VinVL	PAT [18]	60.55
2022	PhoBERT	ResNet-34 + Vision Transformer	MCA [20]	60.76
2023	BARTPho	BEIT-2	BARTPhoBiT [41]	68.58
2024	BARTPho	BLIP2 + EfficientNet-B7	Advancing ViVQA [37]	71.04
2026	PhoBERT	BLIP2 + EfficientNet-B7	**Ours**	**72.15**

Table 2: Comparison of question representation backbones

Question Backbone	Accuracy (%)
BARTPho	68.59
ViT5	69.24
PhoBERT	**72.15**

4.2.4 Comparison of Global Image Representation Backbones

To evaluate the impact of global backbones for image feature extraction, we conduct a comparison among configurations that combine EfficientNet-B7 with different models, including CLIP2, ViT, and BLIP2. The results are presented in Table 3.

The results show that CLIP2 + EfficientNet-B7 and ViT + EfficientNet-B7 achieve relatively lower performance, with accuracies of 61.52% and 60.31%, respectively. In contrast, the BLIP2 + EfficientNet-B7 configuration significantly outperforms the others, achieving the highest accuracy of 72.15% among all compared methods. This indicates that BLIP2 is more effective at capturing semantic visual information when combined with EfficientNet-B7, thereby substantially improving the overall performance of the VQA system.

Table 3: Comparison of global image representation backbones

Visual Extractor	Accuracy (%)
CLIP2 + EfficientNet-B7	61.52
ViT + EfficientNet-B7	60.31
BLIP2 + EfficientNet-B7	**72.15**

To further analyze these results, the superior performance of BLIP2 can be attributed to its strong multimodal architecture design. Specifically, BLIP2 leverages a *Querying Transformer* to effectively bridge visual features and the language representation space, enabling the extraction of more semantically rich features compared to purely visual models such as ViT or contrastive pretraining approaches like CLIP2.

While CLIP2 primarily learns global alignment between images and text through contrastive learning, it tends to overlook fine-grained local details that are crucial for VQA tasks. Similarly, although ViT is effective in modeling spatial relationships, it lacks explicit alignment with linguistic semantics unless it is specifically trained in a multimodal setting.

In contrast, BLIP2 is designed to narrow the gap between vision and language by leveraging large pre-trained language models, resulting in more semantically meaningful visual representations that are better suited for reasoning in VQA tasks. When combined with EfficientNet-B7, the model not only retains strong visual feature extraction capabilities but also enhances multimodal alignment, leading to significant improvements in accuracy.

4.3 Ablation Study

To evaluate the effectiveness of the proposed feature combination strategy in the Hybrid Sequential Fusion module, we conduct a comparative analysis against two simpler alternatives: element-wise addition and concatenation. The results are summarized in Table 4.

Table 4: Ablation study on feature combination strategies.

Feature Combination Strategy	**Accuracy (%)**
$\mathbf{j}_1 + \mathbf{j}_2 + \mathbf{j}_3 + \mathbf{j}_4$	68.04
$Concat(\mathbf{j}_1, \mathbf{j}_2, \mathbf{j}_3, \mathbf{j}_4)$	70.64
Our model (Hybrid Sequential Fusion)	**72.15**

Element-wise addition combines the output features from the four branches $\mathbf{j}_1, \mathbf{j}_2, \mathbf{j}_3, \mathbf{j}_4$ as follows:

$$\mathbf{f}_{add} = \mathbf{j}_1 + \mathbf{j}_2 + \mathbf{j}_3 + \mathbf{j}_4 \tag{35}$$

Concatenation aggregates the feature vectors along the feature dimension:

$$\mathbf{f}_{cat} = [\mathbf{j}_1 \| \mathbf{j}_2 \| \mathbf{j}_3 \| \mathbf{j}_4] \tag{36}$$

As shown in Table 4, element-wise addition achieves an accuracy of 68.04%, while concatenation improves performance to 70.64%. Nevertheless, both alternatives fall short of the proposed model, which attains an accuracy of **72.15%**.

These results demonstrate that naive combination strategies are insufficient for modeling the complex cross-modal interactions inherent in the VQA task.

Element-wise addition, despite preserving the feature dimensionality, fails to capture nonlinear dependencies across branches. Concatenation retains more complete information but lacks any mechanism for filtering or adaptively weighting features according to their relevance. In contrast, the sequential gating mechanism in the Hybrid Sequential Fusion module enables the model to dynamically regulate the contribution of each fusion branch, thereby more effectively exploiting multimodal interactions and yielding superior performance.

5 Conclusion

The experimental results validate the effectiveness of the proposed multimodal fusion strategy, demonstrating consistent improvements over existing baselines on the ViVQA dataset. The dual-branch architecture, comprising the Guided Attention module and the Unified Attention Encoder, captures both fine-grained visual-textual alignment and higher-order semantic dependencies in a complementary manner. The Hybrid Sequential Fusion module further strengthens performance by integrating multiple interaction mechanisms with a sequential gating strategy that adaptively weights each fusion branch, suppressing uninformative features while emphasizing task-relevant representations.

Despite these gains, the model exhibits signs of overfitting, likely due to the limited scale of the ViVQA dataset relative to architectural complexity, and incurs considerable computational overhead from the use of multiple attention and fusion modules. Future work should address these limitations through advanced regularization, data augmentation, large-scale multimodal pretraining, and the development of more parameter-efficient fusion mechanisms.

Overall, this study demonstrates that adaptive attention-based multimodal fusion is an effective paradigm for Vietnamese Visual Question Answering, offering valuable insights for future research on multimodal reasoning in low-resource language settings.

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